Crushing of Soil Particles

By Bobby O. Hardin,[1] F. ASCE

Abstract: In order to understand the physics of the strength and stress-strain behavior of soils and to devise mathematical models that adequately represent such behavior, it is important to define the degree to which the particles of an element of soil are crushed or broken during loading. The amount of particle crushing in a soil element under stress depends on particle size distribution, particle shape, state of effective stress, effective stress path, void ratio, particle hardness, and the presence or absence of water. Data are analyzed for single mineral soils and rockfill-like materials and equations are presented that can be used to estimate the total breakage expected for a given soil subjected to a specified loading.

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Introduction

The strength and stress-strain behavior of an element of soil is affected greatly by the degree to which crushing or particle breakage takes place during loading and deformation. For the type of deformation that primarily produces volume change, such as one-dimensional strain or isotropic compression, particle breakage adds to the reduction in volume. For the type of deformation where particles are moving past or around one another, such as triaxial compression or simple shear, crushing at sliding contacts or breakage of particles decreases the rate of dilation corresponding to a given principal stress ratio. With reference to triaxial and one-dimensional strain testing of rockfill materials, Marsal (19) states the following:

> Apparently, the most important factor affecting both shear strength and compressibility is the phenomenon of fragmentation undergone by a granular body when subjected to changes in its state of stresses, both during the uniform compression stage and during deviator load application. A similar fact is observed in one-dimensional compression tests.

Thus, in order to understand the physics of the strength and stress-strain behavior of soils and to devise mathematical models that adequately represent such behavior, it is important to define the degree to which the particles of an element of soil are crushed or broken. The amount of particle crushing that occurs in an element of soil under stress depends on the following parameters: (1) Particle size distribution; (2) particle shape; (3) state of effective stress; (4) effective stress path; (5) void ratio; (6) particle hardness (hardness of cementing material or weakest constituent of a particle and weakest particles of an element); and (7) the presence or absence of water.

In this paper data are analyzed for soils about which the initial particle

[1]Prof. of Civ. Engrg., Dept. of Civ. Engrg., Univ. of Kentucky, Lexington, Ky.

Note.—Discussion open until March 1, 1986. To extend the closing date one month, a written request must be filed with the ASCE Manager of Journals. The manuscript for this paper was submitted for review and possible publication on October 29, 1984. This paper is part of the *Journal of Geotechnical Engineering*, Vol. 111, No. 10, October, 1985. ©ASCE, ISSN 0733-9410/85/0010-1177/$01.00. Paper No. 20074.

TABLE 1.—Crushing Test Data

Soil name; constituents; reference (1)	Particle shape (2)	e_i (3)	Effective stress paths (4)	$\frac{\sigma'_o}{p_a}$ (5)	$\frac{\tau_o}{p_a}$ (6)	B_p (7)
(*a*) Corundum						
Aluminum oxide sand; Ref. 16	Angular	0.79	TCL	55.3–199	0.643–0.719	0.50
(*b*) Predominantly Quartz Sands						
Mol white quartz sand; Ref. 7		0.64	TCL K_o	48.4–2,005	0.421–0.707	0.41
Chattahoochee river sand, some mica; Refs. 23–25	Subangular	0.72	IC TCL	33.7–1,081	0–0.701	0.67–0.73
Ottawa sand; Refs. 12 and 22	Round	0.49–0.62	TCL K_o	45.3–635	0.354–0.686	0.74–0.99
Ham River sand; Ref. 4		0.82	IC TCL	116–481	0–0.614	0.49
Back Mine quartz sand; Ref. 16	Angular	0.74	IC TCL	15.0–218	0–0.771	0.47
(*c*) Basalt, Epidotized Meta-Volcanic, Argillite, Impure Quartzite						
Napa basalt; Refs. 1, 15 and 17	Angular	0.375–0.698	IC TCL K_o	6.01–94.1	0–1.01	1.43–2.43
Basalt; Ref. 19	Angular	0.30	TCL	50.0	0.730	2.16
Cougar basalt; Ref. 15	Angular	0.492–0.787	TCL K_o	9.36–56.8	0.498–0.895	2.42
New Hogan epidotized meta-volcanic, quartz, epidote, calcite, dolomite, pyrite, albite feldspar; Ref. 15	Angular	0.478–0.781	TCL K_o	8.98–58.4	0.541–0.912	2.41
Pyramid Dam argillite; Ref. 17	Angular	0.451–0.476	TCL PSL	6.63–106	0.691–0.917	1.54–2.15
Carters Dam impure quartzite some muscovite, biotite, calcite, pyrite; Ref. 15	Angular	0.399–0.733	TCL K_o	8.74–57.80	0.555–0.913	2.42
(*d*) Granite, Syenite, Andesite, Amphibolite, Sandstone, Dolomite Limestone, Chert, Sand with Significant Feldspar						
Crushed syenite, feldspar; Ref. 10	Angular	1.02	K_o	28.5	0.555	2.79
Venato sandstone, micaceous well-cemented, fine-medium grained; Ref. 3	Angular	0.433–0.487	TCL PSL	4.77–98.9	0.635–0.810	1.55–2.17
Crushed granite; Ref. 11	Subround to angular	0.58–0.74	R	19.3–124	0–0.530	1.47–1.54
Alluvial andesite (material 5D) rhyolite, meta-volcanics, some granite limestone, quartzite; Ref. 14		0.297	TCL	10.9	0.883	1.71
Ube decomposed granite; Ref. 20	Angular	0.55	TCL	7.66	0.879	0.94
Oroville Dam amphibolite meta-volcanic, mostly augite, some biotite, quartz, schist, diorite, dacite, porphyry, granulite; Ref. 17	Rounded to subround	0.197–0.237	TCL PSL	6.13–118	0.719–0.988	1.32–1.91
Sonora dolomite, some calcite muscovite, pyrite; Ref. 15	Angular	0.327–0.545	TCL K_o	9.62–57.4	0.498–0.856	2.41
Monterey sand, quartz with some feldspar, other minerals; Ref. 17	Subround to round	0.65	TCL PSL	61.4–69.4	0.520–0.631	0.99
Painted Rock gravelly sand quartz, feldspar, chert; Ref. 2	Subround to round	0.394–0.406	IC TCL PSL	10.2–70.2	0–0.886	1.43

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

TABLE 1.—*Continued*

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Alluvial gravel, material 6; Ref. 14		0.287–0.307	IC TCL	17.9–84.3	0–0.797	2.17–2.42
Alluvial graywacke sandstone material 4, some serpentine, diabase; Ref. 14		0.372–0.569	TCL	8.62–9.64	0.745–0.815	0.91–2.32
St. Marc limestone sand, fairly fossiliferous; Ref. 16	Angular	0.81	IC TCL	4.03–227	0–0.818	0.50
(*e*) Weathered Granite, Granitic Gneiss, Argillaceous Sandstone Alluvial Volcanics, Sand Containing Aggregate Particles						
Black Butte Dam alluvial gravel, volcanics, quartzite, jasper, vein quartz, granitic rocks; Ref. 15	Subangular round edges	0.31	K_o	30.5	0.555	2.41
Granitic gneiss, thin layers of schist; Refs. 8, 18 and 19	Angular	0.32–0.62	TCL PSL	3.45–42.8	0.508–0.811	2.17–3.19
Sacramento River sand, feldspar and quartz; Ref. 12 (suspected aggregate particles)	Subangular-subround	0.61	TCL	20.1–238	0.614–0.735	0.43
Antioch sand, 50% quartz, 20% aggregate particles, 10% heavy minerals, 1% mica; Ref. 13	Angular aggregate particles	0.75	TCL	11.6–32.3	0.648–0.867	0.49
Laurel Dam argillaceous sandstone, quartz weakly cemented by secondary silica and kaolinite; Ref. 15	Subangular-angular round edges	0.468–0.819	TCL	8.39–56.4	0.498–0.810	2.42
Buchanan Dam weathered granite, granodiorite, quartz, feldspar, some biotite, chlorite; Ref. 15	Subangular	0.396–0.642	TCL K_o	8.34–61.4	0.471–0.764	2.41
(*f*) Weakly Cemented Aggregate Particles						
Tokyo fine sand and gravel, quartz, feldspar, some hornblende, flaky ferruginous fragments; Ref. 5 (suspected weak calcareous cementation)	Subangular	1.041–1.102	TCL	29.0–1,036	0.490–0.708	0.30

Note: TCL = triaxial compression loading, PSL = plane strain loading, IC = isotropic compression, K_o = one-dimensional strain, and R = constant principal stress ratio loading.

size distribution and the particle size distribution after particle breakage are known. Data have been analyzed for the soils listed in Table 1. The soils in Table 1 are listed in order of decreasing hardness or resistance to particle crushing, as indicated by the analysis. Single mineral soils as well as rockfill-type materials are included. The particle shape and effective stress paths for each soil and the ranges of initial void ratio, e_i, and effective octahedral normal and shear stresses, σ_o' and τ_o, respectively, for the tests on each soil are given in Table 1. The parameter B_p shown in the last column of Table 1 (to be precisely defined later) is related to the particle size distribution.

Previous Measures of Particle Breakage

The amount of particle breakage during loading of a soil sample is defined by the particle size distribution curves measured before and after loading. Several investigators have used a single number or measure to represent the amount of particle breakage that has taken place during

Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.
Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.

loading. The problem is to choose a measure that adequately integrates the breakage occurring in the various size fractions of the material. The increase in percent passing a given sieve size has often been used. As shown in Fig. 1(*a*), Leslie (14) used the percent passing the sieve on which the original material was 100% retained. Later (15) he used the increase in percent passing the sieve on which 90% of the original sample was retained.

For most cases the breakage measure proposed by Marsal (18) falls in the category of increase in percent passing a single sieve size. When the increase in percent passing is stationary at only one diameter over the range of initial particle sizes as in Fig. 1(*a*), Marsal's breakage measure is equal to the increase in percent passing the sieve suffering the greatest increase. However if the increase in percent passing is stationary at more

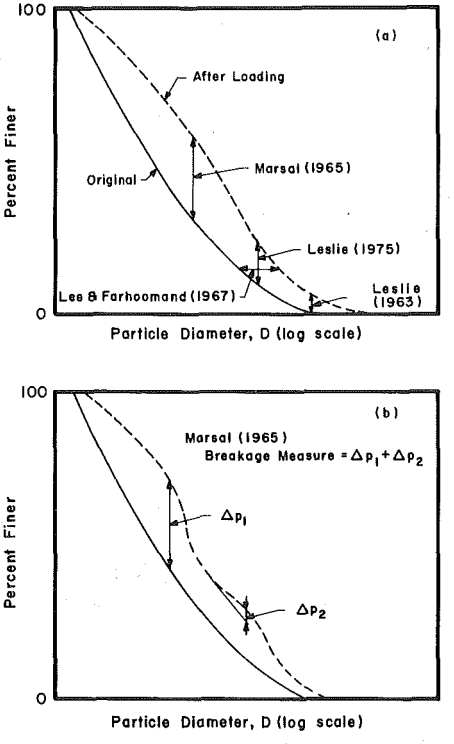


FIG. 1.—Previous Measures of Particle Breakage

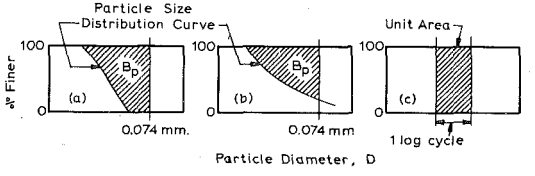


FIG. 2.—Definition of Breakage Potential

than one diameter, Marsal's breakage measure is the sum of more than one difference in percent passing, as shown in Fig. 1(*b*).

The breakage measure can be related to the particle size scale instead of the percent finer scale. Lee and Farhoomand (11) used the ratio D_{15i}/D_{15a} in which D_{15i} = the diameter for which 15% of the original sample is finer; and D_{15a} = the diameter for which 15% of the loaded sample is finer. Since particle size is plotted on a log scale, this ratio is determined by the horizontal distance between particle size distribution curves at 15% finer as shown in Fig. 1(*a*).

Effect of Particle Size Distribution

The potential for breakage of a soil particle increases with its size. This results from the fact that the normal contact forces in a soil element increase with particle size, and the fact that the probability of a defect in a given particle increases with its size. The breakage of soil particles in a sample of rockfill under moderate stresses will be quite evident, whereas very high stresses are required to crush silt size particles. A means of integrating the potential for crushing in an element of soil that contains a range of sizes is needed.

Because of the large stress required to crush silt size particles and because the distribution of silt sizes contained in an element of soil is less important to soil behavior than the distribution of sizes larger than silt, the sizes of particles may be referenced to the largest silt size, 0.074 mm. The purpose of this is to define their potential for "breakage that is significant to soil behavior." The potential for breakage of a particle of a given size, D, may be represented by

$$b_p = \log_{10}\left[\frac{D \text{ in mm}}{0.074 \text{ mm}}\right] \quad \text{for} \quad D \geq 0.074 \text{ mm} \quad \ldots\ldots (1a)$$

$$b_p = 0 \quad \text{for} \quad D < 0.074 \text{ mm} \quad \ldots\ldots (1b)$$

If b_p represents the potential for breakage that is significant to soil behavior, for a given size fraction, df, in an element of soil, where df is a differential of "percent passing" divided by 100, the desired integration of the potential for breakage is given by

$$B_p = \int_0^1 b_p df \quad \ldots\ldots (2)$$

The parameter B_p is called the breakage potential.

As shown in Fig. 2, B_p is equal to the area between the line defining the upper limit of the silt size, $D = 0.074$ mm, and the part of the particle size distribution curve for which $D > 0.074$ mm, where the unit area is the area of one log cycle as shown in Fig. 2(*c*). The value of B_p is easily determined by measuring the hatched areas in Fig. 2 with a planimeter; or if enough points on the particle size distribution curve are digitized, numerical integration may be employed.

Now suppose an element of soil, with breakage potential B_p, is subjected to a state of stress that results in crushing of some particles. The particle size distribution curve after crushing will be shifted as shown

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

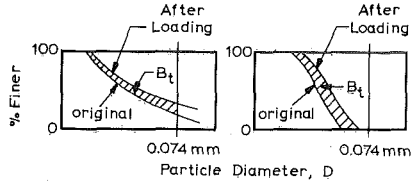


FIG. 3.—Definition of Total Breakage

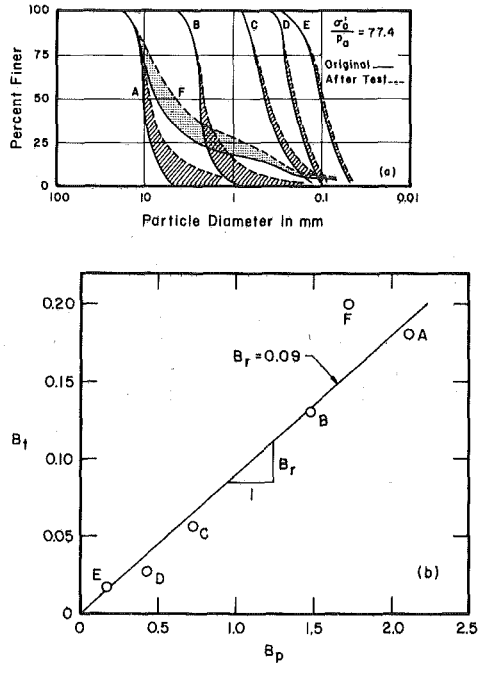


FIG. 4.—(*a*) Particle Size Distribution Curves for Crushed Granite Measured by Lee and Farhoomand (11); (*b*) Relationship between Total Breakage and Breakage Potential

in Fig. 3. The amount of crushing that is significant to soil behavior may be represented by

$$B_t = \int_0^1 (b_{po} - b_{p\ell})\, df \quad \ldots\ldots\ldots\

particle size distribution, B_t is proportional to B_p. This leads to the definition of relative breakage

$$B_r = \frac{B_t}{B_p} \quad \ldots\ldots (4)$$

The relative breakage, B_r, is approximately independent of particle size distribution when particle size distribution is the only variable. This is shown in Fig. 4.

Fig. 4(*a*) shows particle size distributions for a crushed granite measured by Lee and Farhoomand (11) before and after isotropic compression under an effective stress of 80 kg/cm^2. The areas in Fig. 4(*a*) defining B_t and B_p for each set of before and after curves and the unit area have been measured and the values of B_t are plotted versus the corresponding values of B_p in Fig. 4(*b*). The data in Fig. 4(*b*) indicate an approximately linear relationship between B_t and B_p, suggesting that B_r is approximately independent of particle size distribution.

Effect of State of Effective Stress and Effective Stress Path

Under extremely high stresses it would in principle be possible for all of the particles in a sample of soil to be crushed to the extent that no particles remain larger than 0.074 mm. If this happens to a soil element with original breakage potential B_p, then $B_t = B_p$ and $B_r = 1$. Thus, it may be expected that B_r increases with the magnitude of the state of effective stress, approaching $B_r = 1$ for very high stress approximately asymptotically.

The results of measurements of the variation of B_r with state of effective stress are shown in Fig. 5. The data in Fig. 5 are for samples of crushed granite with constant original B_p. The particles are angular and the samples were subjected to constant $R = \sigma_1'/\sigma_3'$ stress paths, in which σ_1' and σ_3' = the major and minor principal effective stresses, respectively. The values of B_r are plotted versus σ_3'/p_a in Fig. 5(*a*) where p_a is atmospheric pressure. Fig. 5(*a*) shows the effect of effective stress path on relative breakage when the minor principal effective stress is used to represent effective stress.

This raises the question as to whether or not it is possible to find a function of the state of effective stress that will adequately define relative breakage in a soil element independent of effective stress path. The breakage effective stress σ_b', defined by Eq. 5 in terms of the effective octahedral normal and shear stresses σ_o' and τ_o, appears to do this:

$$\sigma_b' = \sigma_o'\left[1 + 9\left(\frac{\tau_o}{\sigma_o'}\right)^3\right] \quad \ldots\ldots (5)$$

Eq. 5 was found by trial and error. The values of B_r for the angular crushed granite in Fig. 5(*a*) are plotted versus σ_b'/p_a in Fig. 5(*b*). The data in Fig. 5(*b*) for the three constant R effective stress paths are approximated by a single curve. Other data supporting the use of σ_b' to account for the effects of effective stress path and state of effective stress on crushing will be presented as the analysis proceeds. The relationship between B_r and σ_b' will now be defined.

Downloaded from ascelibrary.org by Saint Consulting Ltd on 10/06/20. Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

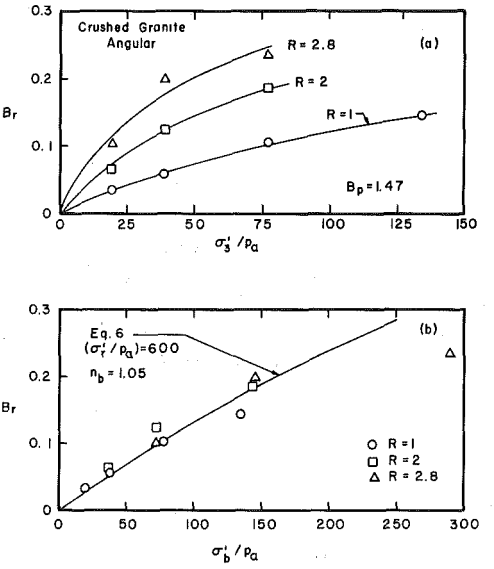


FIG. 5.—Relationship between Relative Breakage and: (*a*) Minor Principal Effective Stress; (*b*) Breakage Effective Stress; Tests by Lee and Farhoomand (11)

Using σ_b' to represent the state of effective stress, the behavior shown in Fig. 5 can be defined by the hyperbolic equation

$$B_r = \frac{\left(\frac{\sigma_b'}{\sigma_r'}\right)^{n_b}}{1 + \left(\frac{\sigma_b'}{\sigma_r'}\right)^{n_b}} \quad \ldots\ldots (6)$$

in which σ_r' = the breakage reference stress; and n_b = the breakage number. When $\sigma_b' = \sigma_r'$, $B_r = 0.5$. The curve in Fig. 5(*b*) is defined by Eq. 5 with $\sigma_r'/p_a = 600$ and $n_b = 1.05$.

As shown by Eq. 8, σ_r' and n_b are related. The values used in Fig. 5(*b*) satisfy Eq. 8. If σ_r' and n_b are assumed to be independent, a somewhat better fit in Fig. 5(*b*) is given by a different set of values.

Eq. 6 can be rearranged to facilitate the determination of σ_r' and n_b. Taking the reciprocal of Eq. 6 and subtracting 1 from both sides gives

$$\left[\frac{1}{B_r} - 1\right] = \left[\frac{\sigma_r'}{\sigma_b'}\right]^{n_b} \quad \ldots\ldots (7)$$

For constant σ_r' and n_b, Eq. 7 defines a straight line on a log-log plot of $[(1/B_r) - 1]$ versus $(1/\sigma_b')$. The arithmetic slope of the line is equal to n_b and σ_r' is equal to the value of σ_b' at $[(1/B_r) - 1] = 1$, where $B_r = 0.5$. Data for five different soils with varying degrees of crushability are plotted in this way in Fig. 6. The four straight lines in Fig. 6 are defined by Eqs. 6 or 7 with n_b = 0.5, 1.0, 1.5, and 2, and σ_r'/p_a = 160, 560, 960, and 1,360, respectively, where

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

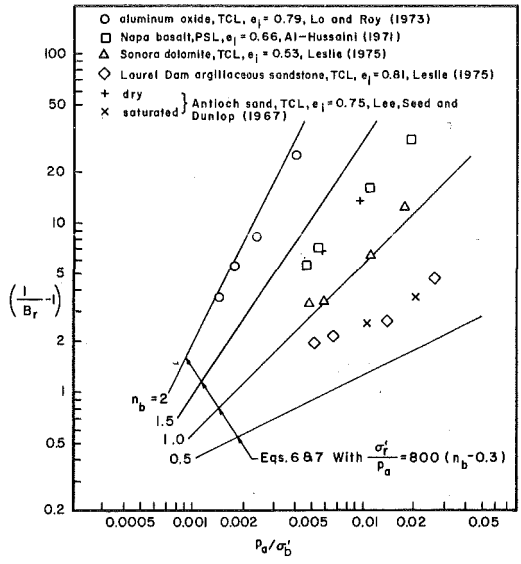


FIG. 6.—Determination of Breakage Reference Stress and Breakage Number

$$\sigma_r' = 800\, p_a(n_b - 0.3) \quad \ldots\ldots\ldots\

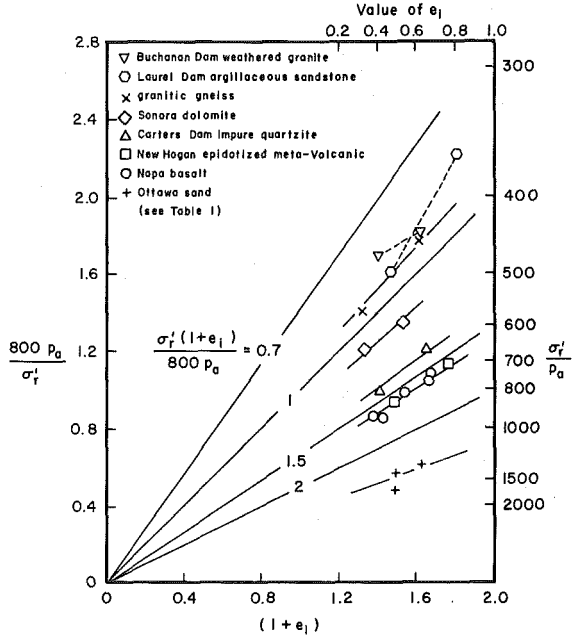


FIG. 7.—Relationship between Breakage Reference Stress and Initial Ratio

(K_o) stress paths and for five values of initial void ratio, ranging from loose to dense, are given in Table 2.

The five sets of average values of e_i and σ_r'/p_a in Table 2 were used to plot the open circle symbols for Napa basalt in Fig. 7. Data for seven additional soils, listed in Table 1, where tests at different initial void ratios were available, are also shown in Fig. 7. The straight lines emanating from the origin in Fig. 7 are defined by $[\sigma_r'(1 + e_i)/(800\ p_a)]$ = constant. Fig. 7 suggests that for the data analyzed

$$\frac{\sigma_r'(1+e_i)}{800\,p_a} = H \quad \ldots\ldots (9)$$

is approximately constant for a given particle shape and hardness. The value of H is readily interpolated from Fig. 7. For example, the average value of H for the angular Napa basalt is 1.59.

Effects of Particle Shape and Hardness

If we define the shape number, n_s, as shown in Table 3, and define the crushing hardness

$$h = (n_s H)^{1/2} \quad \ldots\ldots (10)$$

the test data indicate that crushing hardness, h, is approximately equal to the scratch hardness as defined by Moh's scale. This is illustrated in Table 4.

The computation of σ_r'/p_a was illustrated in Table 2. Once the value

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Downloaded from ascelibrary.org by Stantec Consulting Ltd
Copyright ASCE; all rights reserved, including rights for text and data mining and training of

TABLE 2.—Data for Napa Basalt

Stress path (1)	B_p (2)	b (3)	R (4)	σ_3'/p_a (5)	σ_o'/p_a (6)	τ_o/σ_o' (7)	B_r (%) (8)	σ_b'/p_a (9)	e_i (10)	σ_r'/p_a (11)
(*a*) Tests by Marachi, Chan, Seed and Duncan (17)										
TCL	1.52	0	8.41	2.04	7.08	1.01	2.4	72.1	0.422	923
			5.42	9.52	23.5	0.842	8.9	150	0.431	841
			4.58	28.6	62.7	0.769	17.3	320	0.437	931
			4.39	44.2	94.1	0.750	20.0	452	0.433	1,061
	2.16		6.84	2.04	6.01	0.934	1.3	50.1	0.425	941
			5.10	9.52	22.5	0.817	6.5	133	0.431	885
			4.32	28.6	60.3	0.743	14.4	283	0.438	943
			4.04	44.2	89.0	0.712	19.2	378	0.437	974
Averages: $e_i = 0.432$; $\sigma_r'/p_a = 937$; Std Dev = 64										
(*b*) Tests by Al-Hussaini (1)										
TCL	1.43	0	5.33	4.08	9.97	0.835	5.0	62.3	0.658	722
			4.78	8.50	19.2	0.788	8.6	104	0.658	730
			4.29	20.4	42.8	0.740	16.0	199	0.655	754
			4.11	34.0	69.2	0.720	24.4	302	0.664	752
PSL		0.20	5.86	4.08	12.0	0.713	3.1	51.2	0.669	774
		0.19	5.29	8.50	23.0	0.689	5.8	90.4	0.671	789
		0.18	4.74	20.4	50.4	0.659	12.2	180	0.664	810
		0.16	4.22	28.9	64.9	0.629	15.1	210	0.656	797
Averages: $e_i = 0.662$; $\sigma_r'/p_a = 766$; Std Dev = 32										
TCL	1.43	0	6.29	4.08	11.3	0.902	5.0	85.8	0.533	810
			5.37	8.50	20.9	0.839	9.2	132	0.543	785
			4.62	20.4	45.0	0.773	18.8	232	0.530	754
			4.42	34.0	72.8	0.753	26.8	353	0.525	777
PSL		0.19	7.62	4.08	14.8	0.792	4.4	80.9	0.533	822
		0.20	6.12	8.50	25.9	0.726	7.0	115	0.546	816
		0.18	5.17	20.4	53.9	0.687	12.2	211	0.546	873
		0.18	4.84	28.9	72.6	0.666	16.1	265	0.547	871
Averages: $e_i = 0.538$; $\sigma_r'/p_a = 814$; Std Dev = 42										
(*c*) Tests by Leslie (15)										
TCL	2.41	0	7.40	4.08	12.8	0.963	4.2	115	0.375	948
			5.86	8.50	22.3	0.874	6.4	156	0.375	949
			4.87	20.4	46.7	0.797	14.1	259	0.375	911
			4.66	27.2	60.4	0.777	14.3	315	0.369	1,001
K_o[a]			2.94	18.5	30.5	0.555	4.2	77.4	0.383	820
Averages: $e_i = 0.375$; $\sigma_r'/p_a = 926$; Std Dev = 67										
TCL			5.33	4.08	10.0	0.835	5.8	62.3	0.698	688
			4.54	8.50	18.5	0.765	7.1	93.3	0.673	748
			3.87	20.4	39.9	0.691	14.5	159	0.669	709
			3.81	27.2	52.7	0.684	17.1	204	0.676	741
Averages: $e_i = 0.679$; $\sigma_r'/p_a = 722$; Std Dev = 28										

[a]The value $K_o = 0.34$ used here is based on strength test results.

TABLE 3.—Definition of Shape Number

Particle shape (1)	Value of n_s (2)
Angular	25
Subangular	20
Subround	17
Round	15

of σ_r'/p_a for a given test is determined, σ_r'/p_a and e_i can be substituted into Eq. 9 to calculate the value of H. The value of n_s corresponding to the particle shape can be chosen from Table 3 and used in Eq. 10 with the value of H, to compute the crushing hardness, h.

The results of such computations for h are presented in Table 4 for the soils listed in Table 1. Table 4 shows the number of tests available for each soil, the value of n_s that was used, the average computed value of h, and the standard deviation for h. For those cases where the scratch hardness was reported, the value or range of values is given in Table 4. The scratch hardness values for corundum = 9 and for quartz = 7 are listed in Table 4 in brackets. Comparison of the values of crushing hardness h with the reported or known values of scratch hardness shows that h is approximately equal to the Moh's scale scratch hardness.

Summary of Equations

Combining the equations from the foregoing analysis, the total breakage B_t for a given soil can be expressed as a function of six variables accounting for the effects of particle size distribution, state of effective stress and effective stress path, initial void ratio, particle shape, and particle hardness.

$$B_t = f(B_p, \tau_o, \sigma_o', e_i, n_s, h) \quad \ldots\ldots (11)$$

Substitute $S = \sigma_b'/\sigma_r'$ into Eq. 6, then substitute for B_r in Eq. 4 and solve for B_t. This leads to

$$B_t = \frac{B_p S^{n_b}}{1 + S^{n_b}} \quad \ldots\ldots (12a)$$

Eliminating $\sigma_r'/(800\ p_a)$ from Eqs. 8 and 9 and substituting for H using Eq. 10 gives

$$n_b = \frac{h^2}{(1 + e_i)n_s} + 0.3 \quad \ldots\ldots (12b)$$

Combining Eqs. 5, 9, and 10 leads to

$$S = \frac{(1 + e_i)n_s}{800\ h^2} \frac{\sigma_o'}{p_a} \left[1 + 9 \left\{\frac{\tau_o}{\sigma_o'}\right\}^3\right] \quad \ldots\ldots (12c)$$

Eqs. 12*a–c* can be used to estimate the total breakage B_t expected for a given soil and loading, where h is approximately equal to the scratch hardness and n_s is defined in Table 3.

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.
Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.

TABLE 4.—Crushing Hardness and Scratch Hardness

Material (1)	Shape number, n_s (2)	Number of tests (3)	Crushing hardness, $h \pm$ Std Dev (4)	Reported scratch hardness (5)
(*a*) Corundum				[9]
Aluminum oxide sand	25	4	8.5 ± 0.1	
(*b*) Predominantly Quartz Sands				[7]
Mol white quartz sand	15	6	7.7 ± 1.88	
Chattahoochee River sand	20	10	7.2 ± 2.5	
Ottawa sand	15	6	6.4 ± 0.3	
Ham River sand	19	3	6.2 ± 0.5	
Back Mine quartz	25	7	6.2 ± 0.3	
(*c*) Basalt, Epidotized Meta-Volcanic, Argillite, Impure Quartz				
Napa basalt: Marachi, et al.	25	8	6.5 ± 0.2	
Leslie	25	9	6.2 ± 0.2	6
Al-Hussaini	25	16	6.3 ± 0.2	
Marsal	25	1	6.3	
Cougar basalt	25	9	6.1 ± 0.1	6
New Hogan epidotized meta-volcanic	25	9	6.3 ± 0.2	5.5–6
Pyramid Dam argillite	25	12	6.1 ± 0.6	
Carters Dam impure quartzite	25	9	6.0 ± 0.2	5.5–6
(*d*) Granite, Syenite, Andesite, Amphibolite, Sandstone, Dolomite Limestone, Chert, Sand with Significant Feldspar				
Crushed syenite	25	1	5.9	
Venato sandstone	25	12	5.7 ± 0.9	
Crushed granite	17 and 25	16	5.5 ± 0.4	
Alluvial andesite (material 5D)	17	1	5.5	
Ube decomposed granite	25	1	5.5	
Oroville Dam amphibolite	16	12	5.4 ± 0.6	
Sonora dolomite	25	9	5.3 ± 0.1	5
Monterey sand (quartz with some feldspar and other minerals)	16	3	5.2 ± 0.1	
Painted Rock gravelly sand (quartz, feldspar and chert)	18	9	5.1 ± 0.5	
Alluvial gravel (material 6)	20	8	5.0 ± 0.2	
Alluvial greywacke sandstone (material 4)	20	8	5.0 ± 0.3	
St. Marc limestone sand	25	10	5.0 ± 0.5	
(*e*) Weathered Granite, Granitic Gneiss, Argillaceous Sandstone Alluvial Volcanics, Sands Containing Aggregate Particles				
Black Butte Dam alluvial gravel (volcanics, quartzite, jasper, vein quartz, granitic rocks)	18	1	4.9	
Granitic gneiss	25	10	4.8 ± 0.2	
Sacramento River sand (suspected aggregate particles)	19	4	4.3 ± 0.3	
Antioch sand (20% aggregate particles)	25	2	4.2 ± 0.1	
Laurel Dam argillaceous sandstone	20	9	4.2 ± 0.4	4.5
Buchanan Dam weathered granite	20	9	4.1 ± 0.3	4
(*f*) Suspected Weakly Cemented Aggregate Particles				
Tokyo fine sand and gravel	20	4	2.8 ± 1.6	

Analysis of Breakage of Particles of Given Size

Total breakage is considered in the analysis presented above. In order to understand the details of the particle breakage process, one needs to know the probability of breakage of a particle of a given size, material and shape, in a given environment; and, if the particle breaks, the probability distribution for size of particles resulting from breakage. The author is currently studying this process by considering the initial and final breakage potential quantities, b_{po} and $b_{p\ell}$, to be two dependent random variables.

The joint probability density function is defined by

$$p(b_{p\ell}, b_{po}) = p(b_{p\ell}|b_{po})\, p(b_{po}) \quad \ldots\ldots\ldots\

APPENDIX I.—REFERENCES

1. Al-Hussaini, M. M., "Investigation of Plane Strain Shear Testing. Report 2. Drained Plane Strain and Triaxial Compression Tests on Crushed Napa Basalt," *Report S-71-2*, U.S. Army Engineer Waterways Experiment Station, June, 1971, 120 pp.
2. Al-Hussaini, M. M., "Investigation of Plane Strain Shear Testing. Report 3. Plane Strain and Triaxial Compression Tests on Painted Rock Dam Material," *Report S-71-2*, U.S. Army Engineer Waterways Experiment Station, Sept., 1972, 132 pp.
3. Becker, E., Chan, C. K., and Seed, H. B., "Strength and Deformation Characteristics of Rockfill Materials in Plane Strain and Triaxial Compression Tests," *Report No. TE 72-3*, University of California, Berkeley, Oct., 1972, 121 pp.
4. Bishop, A. W., "The Strength of Soils as Engineering Materials," *Geotechnique*, Vol. 16, No. 2, June, 1966, pp. 91–128.
5. BCP Committee, "Field Tests on Piles in Sand," *Soils and Foundations*, JSSMFE, Vol. 11, No. 2, June, 1971, pp. 29–49.
6. Bird, J. M., "Uncertainties in Earth Dam Design," *Journal of the Soil Mechanics and Foundations Division*, ASCE, Vol. 87, No. SM3, June, 1961, pp. 33–68.
7. deBeer, E. E., "The Scale Effect in the Transposition of the Results of Deep-Sounding Tests on the Ultimate Bearing Capacity of Piles and Caisson Foundations," *Geotechnique*, Vol. 13, No. 1, Mar., 1963, pp. 39–75.
8. Hillis, S. F., discussion of "Compressibility and Crushing of Granular Soil in Anisotropic Triaxial Compression," *Canadian Geotechnical Journal*, Vol. 4, No. 1, Feb., 1967, pp. 90–92.
9. Holestol, K., Kjaernsli, B., and Torblaa, I., "Compression of Tunnel Spoil at Venemo Dam," *Proceedings of the 6th International Conference on Soil Mechanics and Foundation Engineering*, Montreal, Vol. 2, 1965, pp. 490–494.
10. Kjaernsli, B., and Sande, A., "Compressibility of Some Coarse-Grained Materials," *Proceedings of the European Conference on Soil Mechanics and Foundation Engineering*, Weisbaden, 1963, pp. 245–251.
11. Lee, K. L., and Farhoomand, I., "Compressibility and Crushing of Granular Soil," *Canadian Geotechnical Journal*, Vol. 4, No. 1, Feb., 1967, pp. 68–86.
12. Lee, K. L., and Seed, H. B., "Drained Strength Characteristics of Sands," *Journal of the Soil Mechanics and Foundations Division*, ASCE, Vol. 93, No. SM6, Nov., 1967, pp. 117–141.
13. Lee, K. L., Seed, H. B., and Dunlop, P., "Effect of Moisture on the Strength of a Clean Sand," *Journal of the Soil Mechanics and Foundations Division*, ASCE, Vol. 93, No. SM6, Nov., 1967, pp. 17–40.
14. Leslie, D. D., "Large Scale Triaxial Tests on Gravelly Soils," *Proceedings of the 2nd Panamerican Conference on Soil Mechanics and Foundation Engineering*, Brazil, Vol. 1, 1963, pp. 181–202.
15. Leslie, D. D., "Shear Strength of Rockfill," *Physical Properties Engineering Study No. 526*, South Pacific Division, Corps of Engineers Laboratory, Sausalito, Calif., Oct., 1975, p. 124.
16. Lo, K. Y., and Roy, M., "Response of Particulate Materials at High Pressure," *Soils and Foundations*, JSSMFE, Vol. 13, No. 1, Mar., 1973, pp. 61–76.
17. Marachi, N. D., Chan, C. K., Seed, H. B., and Duncan, J. M., "Strength and Deformation Characteristics of Rockfill Materials," *Report No. TE-69-5*, University of California, Berkeley, Sept., 1969, 139 pp.
18. Marsal, R. J., "Discussion of Shear Strength," *Proceedings of the 6th International Conference on Soil Mechanics and Foundation Engineering*, Montreal, Vol. 3, 1965, pp. 310–316.
19. Marsal, R. J., "Large Scale Testing of Rockfill Materials," *Journal of the Soil Mechanics and Foundations Division*, ASCE, Vol. 93, No. SM2, Mar., 1967, pp. 27–43.
20. Miura, N., and O-hara, S., "Particle Crushing of a Decomposed Granite Soil Under Shear Stress," *Soils and Foundations*, JSSMFE, Vol. 19, No. 3, Sept., 1979, pp. 1–14.

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

21. Ramamurthy, T., Kanitkar, V. K., Prakash, K., and Chawla, C., "Stress-Strain Relationship for Granular Soils at Low and High Confining Stresses," *Proceedings of the 8th International Conference on Soil Mechanics and Foundation Engineering*, Moscow, Vol. 4.3, 1973, pp. 168–171.
22. Roberts, J. E., and DeSouza, J. M., "The Compressibility of Sands," *Proceedings*, ASTM, Vol. 58, 1958, pp. 1269–1277.
23. Vesic, A. S., and Barksdale, R. D., "Discussion," *Laboratory Shear Testing of Soils*, ASTM, STP 361, 1963, pp. 301–305.
24. Vesic, A. S., "Discussion of Shear Strength," *Proceedings of the 6th International Conference on Soil Mechanics and Foundation Engineering*, Montreal, Vol. 3, 1965, pp. 350–352.
25. Vesic, A. S., and Clough, G. W., "Behavior of Granular Materials Under High Stresses," *Journal of the Soil Mechanics and Foundations Division*, ASCE, Vol. 94, No. SM3, May, 1968, pp. 661–688.

Downloaded from ascelibrary.org by Stantec Consulting Ltd on 10/06/26.
Copyright ASCE; all rights reserved, including rights for text and data mining and training of artificial intelligence or similar technologies.

APPENDIX II.—NOTATION

The following symbols are used in this paper:

B_p = breakage potential, Eq. 2;
B_r = relative breakage, Eq. 4;
B_t = total breakage, Eq. 3;
b_p = potential for breakage of given size fraction, Eqs. 1;
$b_{p\ell}$ = value of b_p after loading;
b_{po} = value of b_p before loading;
D = particle diameter;
D_{15i}, D_{15a} = diameter for which 15% is finer before and after loading, respectively;
e_i = void ratio;
H = parameter defined by Eq. 9;
h = crushing hardness, approximately equal to Moh's scratch hardness;
(K_o) = one-dimensional strain loading;
n_b = breakage number, Eq. 6;
n_s = shape number, Table 3;
(PSL) = plane strain loading;
$p(\ )$ = probability;
p_a = atmospheric pressure;
R = principal stress ratio;
S = parameter defined by Eq. 12c;
(TCL) = triaxial compression loading;
σ_1', σ_3' = major and minor principal stresses, respectively;
σ_b' = breakage effective stress, Eq. 5;
σ_o' = octahedral normal stress;
σ_r' = breakage reference stress, Eq. 6; and
τ_o = octahedral shear stress.